

Dual Trajectory Revised Diffusion Model for Time Series Forecasting

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Abstract—Diffusion models have exhibited state-of-the-art performance in generative tasks across various domains. A few recent works leveraged the powerful modeling ability of the diffusion model to time-series forecasting, leading to a significant breakthrough. However, all these works perform the forecasting through incorporating the historical time-series conditions into the backward denoising. This causes the diffusion model to lose the essential consistency between forward and backward processes, thereby limiting the precision of the inference. In this paper, we propose a novel Dual Trajectory Revised Diffusion Model (TimeDTR) for time-series forecasting, which leverages an unconventional conditioning strategy to incorporate the historical information into both forward and backward trajectories in the diffusion model. Experimental results on six real-world datasets demonstrate that TimeDTR takes a big step forward from the state-of-the-art in time-series forecasting, especially in the long-term forecasting tasks, in terms of forecasting accuracy. The codes of the experiments with datasets and our algorithms are available at <https://github.com/hhzzlll/TimeDTR>.

Index Terms—Time series forecasting, diffusion model, conditional diffusion, trajectory revision.

I. INTRODUCTION

Time series forecasting is a crucial task that addresses numerous real-life challenges by identifying underlying patterns in time series data and extrapolating them to future data points. The predictions derived from such analyses find broad applications across different domains, including finance [1], transportation [2], stock market [3], and energy [4].

Currently, numerous scholars have conducted extensive research in time series forecasting. Among them, methods that based on diffusion models have achieved significant breakthroughs in recent years [5]–[7]. Existing diffusion-based forecasting methods can be roughly classified into two groups: autoregressive methods where the forecasting results are generated based on the last observations [7], and non-autoregressive methods which first learn the distributions of the time series and then generate the forecasting results through conditionally sampling [5], [6], [8]. Although the autoregressive methods perform well in short-range forecasting, their performance in long-range forecasting is relatively poor due to the accumulated error. Non-autoregressive methods can address the

long-term forecasting problem well. However, most existing diffusion-based non-autoregressive methods face one (or more) following issues that prevent them from expected forecasting performance [9]. **Firstly**, as the conditioning strategies are initially designed for images, the sampling results are inferior to time-series data. **Secondly**, most of them perform the conditioning by introducing inductive bias into the denoising process, which may be ineffective in capturing sufficient information from the condition for forecasting. **Lastly**, these methods employ an unconditional forward process to perturb the time series, whereas generating the future series in the denoising process conditionally with the historical data. This inconsistency between forward and backward processes may restrict the forecasting performance of the diffusion models.

To tackle the above issues, we propose a novel diffusion-based forecasting method for time series, named **Dual Trajectory Revised Diffusion Model (TimeDTR)**. It applies an innovative conditioning strategy, which not only fuses the forecasting information with the historical sequence to guide the model to sufficiently exploit the conditional information for forecasting, but also modifies the forward and the backward trajectories together to ensure the consistency of the perturbing and denoising process on time series samples. Besides, we developed a Feature Extraction Adapter (FEA) module and a Past-Future Mapping (PFM) module, which are tailor-made for time series forecasting, to assist the diffusion model in capturing the critical temporal properties in the time-series condition. Empirical results revealed that, among 15 recent benchmarks, our TimeDTR achieves the top-rank performance over six real-world datasets. In particular, it improves the accuracy of the state-of-the-art by 21.2% in long-term forecasting tasks.

In summary, this paper makes the following contributions:

- (1) We propose a novel diffusion-based time-series forecasting model (TimeDTR), which makes meaningful improvements over the state-of-the-art methods, especially in long-term forecasting.
- (2) We design a new conditioning strategy tailored to time-series forecasting, which improves the sampling accuracy of the diffusion models under conditions.

II. RELATED WORK

Time Series Forecasting: Typical time-series forecasting methods include linear-based methods like DLinear [10],

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RNN-based methods like HPRNN [11], and transformers-based methods like FedFormer [12]. Recently, the diffusion models, with powerful generative capabilities, have enabled significant breakthroughs in time-series forecasting [5], [6], [13].

Conditional Diffusion Strategies: TimeGrad [7] is a conditional diffusion model which predicts in an autoregressive manner. CSDI [5] uses non-autoregressive generation and self-supervised masking to guide the denoising process. SSSD [6] reduces the computational complexity of CSDI by replacing the transformers with a structured state space model. TimeDiff [8] uses a non-autoregressive strategy for conditionally sampling. Mr-Diff [14] decomposes the time series into multiple resolutions using seasonal trend decomposition to guide the denoising process.

Diffusion Trajectory Optimization: In recent years, diffusion trajectory optimization has attracted increasing attentions in the image area. GradTTS [15] optimizes the forward process with data-dependent priors for audio diffusion models. CON-TEXTDIFF [16] utilizes a context-aware trajectory adapter to contextualize the diffusion and sampling processes. However, there is no relevant literature on time series forecasting to the best of our knowledge.

III. PROPOSED TIMEDTR

A. Problem Formulation

For the task of time series forecasting, we segment a portion of time series into historical window series $x_{1:L}^0 \in \mathbb{R}^{d \times L}$ and forecast window series $y_{1:H}^0 \in \mathbb{R}^{d \times H}$. Here, L is the length of the historical window, H is the length of the forecast window, and d is the dimension of the time series. A time-series is classified as univariate when $d = 1$ and as multivariate when $d > 1$. Our goal is to forecast the time series in the forecast window given the time series in the historical window.

B. Model Overview

Fig.1 presents the framework of TimeDTR. In the forward diffusion process, time series $y_{1:H}^0$ in the forecast window is perturbed by the revised diffusion trajectory, incorporating a past-future-fused feature $\mathcal{C}_t$ extracted by the Feature Extraction Adapter (FEA). In the backward denoising process, the noisy time series $y_{1:H}^T$ is inputted into the denoising network, and denoised step by step along the revised denoising trajectory to achieve the forecasting result $\hat{y}_{1:H}^0$. To ensure the consistency of the bi-directional trajectory, the denoising process should incorporate the past-future-fused feature as well. Hence, we design a Past Future Mapping (PFM) module that uses a trainable LSTM-based network to predict a primary future time series $\tilde{y}_{1:H}^0$, which is concatenated with $x_{1:L}^0$ to produce the fused feature. After training, TimeDTR can perform an accuracy forecasting using the denoising process. Next, we elaborate our critical designs in TimeDTR, including the revised trajectory, the FEA module, and the PFM module.

C. Revised Diffusion Trajectory

In order to ensure the consistency of the forward diffusion and the backward denoising conditioning on the historical time series, we revise the diffusion trajectories of the original diffusion model, inspired by the text-to-image diffusion model in [16]. Specifically, we first concatenate the time series in historical and forecast windows, and then extract the key condition (denoted by $\mathcal{C}$) for forecasting by FEA. Finally, we propagate this information as a bias term throughout both the forward diffusion steps and the backward denoising steps.

For the forward diffusion process, we add the forecasting condition extracted by FEA to the mean value of the transition probability by:

$$q(y_{1:H}^t | y_{1:H}^0) = \mathcal{N}(y_{1:H}^t, \sqrt{\alpha_t} y_{1:H}^0 + \lambda_t \mathcal{C}_t, (1 - \alpha_t) \mathbf{I}), \quad (1)$$

where $\alpha_s \in [0, 1]$ denotes the weight that controls how much noise is added, $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, $\mathbf{I} \in \mathbb{R}^{d \times H}$ is an identity matrix. λ_t is the weight of the integrated feature. In practice, λ_t is set to $\sqrt{\alpha_t} \cdot (1 - \sqrt{\alpha_t})$ to achieve the optimal performance.

According to (1), the probability of the transition probability from $y_{1:H}^{t-1}$ to $y_{1:H}^t$ can be revised by:

$$q(y_{1:H}^t | y_{1:H}^{t-1}, y_{1:H}^0) = \mathcal{N}(\sqrt{\alpha_t} y_{1:H}^{t-1} + \lambda_t \mathcal{C}_t - \sqrt{\alpha_t} \lambda_{t-1} \mathcal{C}_{t-1}, \beta_t \mathbf{I}), \quad (2)$$

where $\beta_t = 1 - \alpha_t$, is the variance of the noise that generates $y_{1:H}^t$ by corrupting $y_{1:H}^{t-1}$.

For the backward denoising process, the conditional sampling is then modified by (3) to align with the forward diffusion process.

$$q(y_{1:H}^{t-1} | y_{1:H}^t, y_{1:H}^0) = \mathcal{N}\left(\frac{\sqrt{\alpha_{t-1}} \beta_t}{1 - \alpha_t} y_{1:H}^t, t\right) + \frac{\sqrt{\alpha_t} (1 - \alpha_{t-1})}{1 - \alpha_t} (y_{1:H}^t - \lambda_t \mathcal{C}_t) + \lambda_{t-1} \mathcal{C}_{t-1}, \frac{(1 - \alpha_{t-1}) \beta_t}{1 - \alpha_t} \mathbf{I}), \quad (3)$$

where $y_{1:H}^t$ is estimated by the Denoising network by inputting $y_{1:H}^t$ and diffusion step t . Due to space limits, we omit the theoretical proof of the consistency between (1) and (3), which can be concluded from [16].

In the training phase, for each $y_{1:H}^0$, we first randomly sample a batch of diffusion steps, and then minimize a conditioned variant $\min_{\theta} L(\theta) = \min_{\theta} \mathbb{E}_{y_{1:H}^0, \epsilon \sim \mathcal{N}(0, \mathbf{I}), t} L_t(\theta)$, where

$$L_t(\theta) = \|y_{1:H}^0 - y_{\theta}(y_{1:H}^t, t)\|_2^2. \quad (4)$$

Since time-series data usually exhibits irregularities, we let the Denoising network predict the time-series sample $y_{\theta}(y_{1:H}^t, t)$, rather than predicting the noise $\epsilon_{\theta}(y_{1:H}^t, t)$ in typical diffusion models to achieve better performance.

In the inference phase, we first generate a noise vector $y_{1:H}^T$ with Gaussian distribution of size $H \times d$. Then, the forecasting result can be obtained by repeatedly conducting the denoising step with (3). Notably, the ground truth of the future time series is unknown in the inference phase. In such a case, we apply a trainable module PFM (refer to Sec. III-E) to estimate

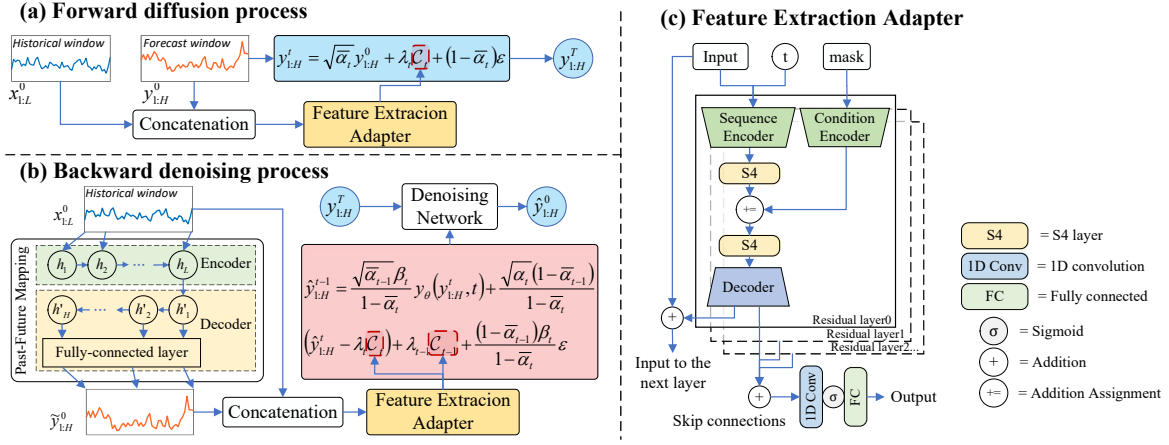


Fig. 1. Overview of TimeDTR Framework: the left is the revised forward and backward processes of the diffusion model, utilizing PFM and FEA to incorporate the forecasting conditions. The right is the structure of the FEA module customized for producing the condition for time-series forecasting.

the future time series given the historical sequence, rather than directly using the predicted $y_\theta(y_{1:H}^t, t)$ to produce the integrated feature C_t . The experimental results in Sec. IV-B will demonstrate the positive effects of PFM.

D. Feature Extraction Adapter

The FEA module is designed to capture the key temporal feature for forecasting from the combined time series in the historical window and forecast window. The FEA module contains several residual layers, allowing for the extraction of more detailed information. We employ a masking strategy in which the elements of the historical observation are set to 1 and the remaining are set to 0. This specific masking strategy is more in line with our forecasting task compared to random masking, ensuring that the information of historical window can be sufficiently exploited for forecasting the remaining series.

As shown in Fig.1, each residual layer of FEA is composed of two Encoders, two Structured State Space Sequences (S4s) and one Decoder. The Sequence Encoder is responsible for embedding the combined time series with the diffusion step t , while the Condition Encoder is used to extract the feature from the masked time series. After the two Encoders, two S4 layers are employed to process the temporal dependencies of the time series. The first S4 captures the long-term dependencies from the combined sequence, while the second S4 models the dependencies when the conditional information is added along with the input. The result is then processed through a Decoder to generate the output of each residual layer. The final output of FEA is activated by a Sigmoid function and mapped by a fully connected layer after being processed by a 1D convolutional layer. Hence, the condition feature C_t for the revised trajectory can be formulated by:

$$C_t = \text{FEA}(\text{concat}(x_{1:L}^0, y_{1:H}^0), t) \quad (5)$$

Where $x_{1:L}^0$ is the time series in the historical window, $y_{1:H}^0$ (or $\tilde{y}_{1:H}^0$) is the corresponding series in the forecast window.

TABLE I
DATASET STATISTICS: INCLUDING DIMENSION (DIM), NUMBER OF OBSERVATIONS (#OBS), SAMPLING FREQUENCY (FREQ), AND FORECASTING LENGTH IN EXPERIMENTS (H).

Dataset	Dim	#Obs	Freq	H (steps)
Weather	21	52696	10 mins	1 weeks(672)
ETTm1	7	69680	15 mins	2 days(192)
Wind	7	48673	15 mins	2 days(192)
ETTh1	7	17420	1 hour	1 week(168)
Electricity	321	26304	1 hour	1 week(168)
Exchange	8	7588	1 day	2 weeks(14)

E. Past-Future Mapping

To ensure the consistency of the condition feature C_t in the two-way trajectory, we design a PFM module to estimate an approximate forecast window from each historical window for the denoising process.

PFM is composed of an Encoder and a Decoder, where the Encoder contains several LSTM layers, and the Decoder contains several LSTM layers and one fully-connected layer. The time series in the historical window are firstly fed into the Encoder, which maps them into hidden states in a recurrent manner:

$$h_n = \text{LSTM}_{en}(h_{n-1}, x_n), \quad n = \{1, 2, \dots, L\}, \quad (6)$$

where x_n is the input time series, and h_n is the hidden state.

Afterwards, the hidden states are decoded into the predictions of the time series in the forecast window iteratively through the LSTM layers and the fully connected layer in the Decoder:

$$h'_m = \text{LSTM}_{de}(h'_{m-1}, \tilde{y}_{m-1}), \quad (7)$$

$$\tilde{y}_m = \mathbf{W}h'_m + \mathbf{b}, \quad m = \{1, 2, \dots, H\}, \quad (8)$$

where h'_m is the hidden state, and $\tilde{y}_m$ is the prediction of the time series. $\mathbf{W}$ and $\mathbf{b}$ are the weights and biases of the fully connected layer. Note that $h'_0 = h_L$ and $\tilde{y}_0 = x_L$.

TABLE II

OVERALL PERFORMANCE COMPARISON: NUMBER IN BRACKETS IS THE PERFORMANCE RANK. THE BEST RESULTS FOR EACH DATASET ARE IN BOLD, WHILE THE SECOND BESTS ARE UNDERLINED. CSDI RUNS OUT OF MEMORY ON ELECTRICITY. RESULTS OF ALL BASELINES (EXCEPT TIMEDIFF) ARE FROM [14].

	ETTh1		ETTm1		Exchange		Weather		Wind		Electricity		Avg rank	
	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
TimeDTR	0.414 ₍₁₎	<u>0.412</u> ₍₂₎	0.365 ₍₁₎	0.330 ₍₁₎	0.088 ₍₃₎	<u>0.017</u> ₍₂₎	0.297 ₍₁₎	0.234 ₍₁₎	<u>0.706</u> ₍₂₎	<u>0.967</u> ₍₂₎	<u>0.304</u> ₍₂₎	<u>0.193</u> ₍₂₎	1.7	1.7
TimeDiff [8]	0.416 ₍₂₎	0.411 ₍₁₎	0.370 ₍₂₎	0.340 ₍₂₎	0.088 ₍₃₎	0.018 ₍₄₎	0.312 ₍₂₎	0.297 ₍₂₎	0.725 ₍₅₎	1.005 ₍₄₎	0.315 ₍₅₎	0.201 ₍₄₎	<u>3.2</u>	<u>2.8</u>
TimeGrad [7]	0.719 ₍₁₆₎	0.993 ₍₁₆₎	0.605 ₍₁₆₎	0.874 ₍₁₆₎	0.193 ₍₁₃₎	0.079 ₍₁₄₎	0.381 ₍₁₁₎	0.392 ₍₁₁₎	0.793 ₍₁₅₎	1.209 ₍₁₅₎	0.630 ₍₁₅₎	0.736 ₍₁₅₎	14.3	14.5
CSDI [5]	0.438 ₍₄₎	0.497 ₍₅₎	0.442 ₍₁₂₎	0.529 ₍₁₃₎	0.194 ₍₁₄₎	0.077 ₍₁₃₎	0.374 ₍₉₎	0.356 ₍₇₎	0.741 ₍₈₎	1.066 ₍₇₎	-	-	9.4	9
SSSD [6]	0.561 ₍₁₃₎	0.726 ₍₁₄₎	0.406 ₍₈₎	0.464 ₍₁₀₎	0.127 ₍₈₎	0.061 ₍₁₁₎	0.350 ₍₆₎	0.349 ₍₆₎	0.778 ₍₁₃₎	1.188 ₍₁₄₎	0.363 ₍₁₀₎	0.255 ₍₁₀₎	9.7	10.8
D ³ VAE [17]	0.502 ₍₁₂₎	0.504 ₍₇₎	0.391 ₍₆₎	0.362 ₍₆₎	0.301 ₍₁₆₎	0.200 ₍₁₆₎	0.380 ₍₁₀₎	0.375 ₍₉₎	0.779 ₍₁₄₎	1.118 ₍₁₂₎	0.372 ₍₁₁₎	0.286 ₍₁₂₎	11.5	10.3
FILM [18]	0.436 ₍₃₎	0.426 ₍₄₎	0.374 ₍₃₎	0.347 ₍₄₎	0.079 ₍₁₎	0.016 ₍₁₎	0.336 ₍₄₎	0.327 ₍₄₎	0.717 ₍₃₎	0.984 ₍₃₎	0.320 ₍₆₎	0.210 ₍₆₎	3.3	3.7
Depts [19]	0.491 ₍₁₀₎	0.579 ₍₁₁₎	0.412 ₍₉₎	0.380 ₍₈₎	0.100 ₍₅₎	0.020 ₍₅₎	0.394 ₍₁₃₎	0.761 ₍₁₅₎	0.751 ₍₉₎	1.082 ₍₉₎	0.401 ₍₁₄₎	0.319 ₍₁₄₎	10	10.3
N-Hits [20]	0.480 ₍₆₎	0.498 ₍₆₎	0.388 ₍₅₎	0.353 ₍₅₎	<u>0.085</u> ₍₂₎	<u>0.017</u> ₍₂₎	0.335 ₍₃₎	0.323 ₍₃₎	0.734 ₍₆₎	1.033 ₍₅₎	0.245 ₍₁₎	0.152 ₍₁₎	3.8	3.7
PatchTST ⁶ [21]	0.489 ₍₉₎	0.526 ₍₉₎	0.392 ₍₇₎	0.372 ₍₇₎	0.147 ₍₁₀₎	0.047 ₍₉₎	0.555 ₍₁₆₎	0.782 ₍₁₆₎	0.720 ₍₄₎	1.070 ₍₈₎	0.348 ₍₉₎	0.225 ₍₈₎	9.2	9.5
FedFormer [12]	0.484 ₍₈₎	0.541 ₍₁₀₎	0.413 ₍₁₀₎	0.426 ₍₉₎	0.233 ₍₁₅₎	0.133 ₍₁₅₎	0.347 ₍₅₎	0.342 ₍₅₎	0.762 ₍₁₁₎	1.113 ₍₁₁₎	0.341 ₍₈₎	0.238 ₍₉₎	9.5	9.8
Autoformer [22]	0.484 ₍₈₎	0.516 ₍₈₎	0.496 ₍₁₄₎	0.565 ₍₁₄₎	0.167 ₍₁₁₎	0.056 ₍₁₀₎	0.354 ₍₇₎	0.360 ₍₈₎	0.756 ₍₁₀₎	1.083 ₍₁₀₎	0.313 ₍₄₎	0.201 ₍₄₎	9	9
Pyraformer [23]	0.493 ₍₁₁₎	0.579 ₍₁₁₎	0.435 ₍₁₁₎	0.493 ₍₁₂₎	0.112 ₍₇₎	0.032 ₍₇₎	0.385 ₍₁₂₎	0.394 ₍₁₂₎	0.735 ₍₇₎	1.061 ₍₆₎	0.379 ₍₁₂₎	0.273 ₍₁₁₎	10	9.8
Informer [25]	0.605 ₍₁₅₎	0.775 ₍₁₅₎	0.542 ₍₁₅₎	0.673 ₍₁₅₎	0.192 ₍₁₂₎	0.073 ₍₁₂₎	0.364 ₍₈₎	0.385 ₍₁₀₎	0.772 ₍₁₂₎	1.168 ₍₁₃₎	0.383 ₍₁₃₎	0.298 ₍₁₃₎	12.5	13
Scaleformer [24]	0.579 ₍₁₄₎	0.613 ₍₁₃₎	0.475 ₍₁₃₎	0.481 ₍₁₁₎	0.138 ₍₉₎	0.036 ₍₈₎	0.438 ₍₁₄₎	0.462 ₍₁₃₎	0.864 ₍₁₆₎	1.359 ₍₁₆₎	<u>0.304</u> ₍₂₎	0.195 ₍₃₎	11.3	10.7
DLinear [10]	0.442 ₍₅₎	0.415 ₍₃₎	0.378 ₍₄₎	0.345 ₍₃₎	0.102 ₍₆₎	0.022 ₍₆₎	0.444 ₍₁₅₎	0.488 ₍₁₄₎	0.686 ₍₁₎	0.899 ₍₁₎	0.336 ₍₇₎	0.215 ₍₇₎	6.3	5.7

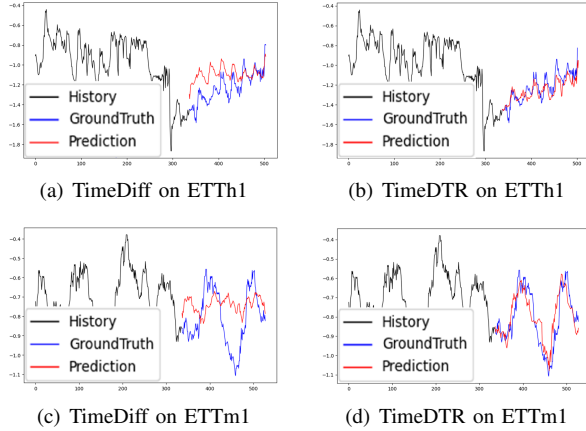


Fig. 2. Visualizations of Forecasting Performance.

IV. EXPERIMENTS

A. Datasets and Baselines

We evaluate TimeDTR on six real-world datasets (as shown in Table I): (i)-(ii) ETTh1 and ETTm1 [24], (iii) Exchange [26], (iv) Weather, (v) Wind [17], and (vi) Electricity.

We compare TimeDTR with 15 benchmark methods grouped into four categories: (i) diffusion models for time series tasks, including TimeDiff [8], TimeGrad [7], CSDI [5], SSSD [6], and D³VAE [17]; (ii) recent forecasting models based on basis expansion, including FiLM [18], Depts [19] and N-Hits [20]; (iii) Transformers-based time-series forecasting methods, including PatchTST⁶ [21], Fedformer [12], Autoformer [22], Pyraformer [23], Informer [24] and Scaleformer [25]; and (iv) Linear-based methods, including DLinear [10].

B. Experimental Results and Analysis

The experimental results are reported in Table II. From the results, we can observe that TimeDTR shows the best overall performance. The average rank of TimeDTR reaches the first over all baselines, in both MAE and MSE. Notably, TimeDTR

TABLE III
ABLATION STUDY OF TIMEDTR.

PFM	FEA	ETTm1	Weather	Electricity
×	✓	0.590	0.304	0.239
✓	×	0.340	0.297	0.201
✓	✓	0.330	0.261	0.198

shows significant improvement in long-term forecasting, such as Weather (with 672 forecasting steps) and ETTm1 (with 192 forecasting steps). This owns to the customized diffusion trajectories that incorporate the temporal features for forecasting using FEA and PFM.

Fig.2 visualizes the forecasting performance of TimeDTR and Timediff (the baseline proposed in 2024, which has the suboptimal performance in Table II) on ETTh1 and ETTm1. As can be seen, with the same historical data, the forecasting curve of TimeDTR clearly fits the ground-truth more closely.

Table III lists the ablation results of FEA and PFM. We first remove PFM from TimeDTR, and combine the historical window with $y_\theta(y_{1:H}^t, t)$ predicted by the Denoising network in the backward process (the first line in Table III). Then we remove FEA from TimeDTR, and apply the typical forward and backward process in diffusion model conditioning on the combined series with PFM (the second line in Table III). The results demonstrate that both PFM and FEA significantly enhance the model's performance.

V. CONCLUSION

In this paper, we propose TimeDTR, the first trajectory-revised diffusion model for time-series forecasting. Unlike existing diffusion-based time-series forecasting models, we incorporate a feature condition customized for time-series forecasting to both the forward and backward process of diffusion models, prompting nontrivial improvements over the state-of-the-art methods in terms of forecasting accuracy. Experimental results on six real-world datasets demonstrate the superiority of TimeDTR.

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